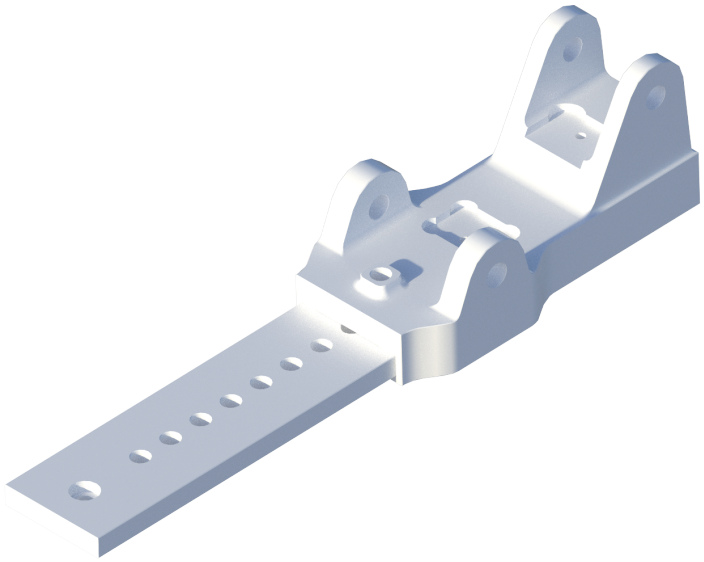
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# Brake Pedal Mounting Analysis



**MSc in Aeronautic and Vehicular Structures**
**Finite Element Method**

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As seen in figure 2.1 – since the force exerted by the master cylinders can be deduced – to determine the reaction in the pivot point, only the sum of the forces on each axis will be needed.

$$\begin{cases} R_X = sin(12°) \cdot 7420 - 2000 = 513.1N \\ R_Y = cos(12°) \cdot 7420 = -7269.5N \end{cases} \quad (2.2)$$

**Adjustable Base**

By transferring the calculated forces to the adjustable base, the following free body diagram is obtained:

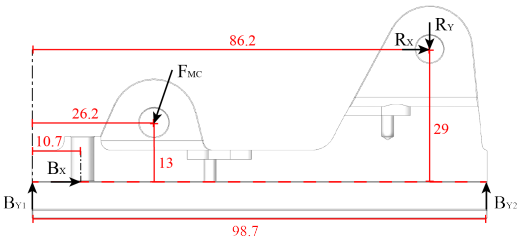



Figure 2.2: Adjustable base free body diagram

Considering the system to be static, the following expression can be obtained by determining the moment created at the edge of the part:

$$-26.2 \cdot cos(12°) \cdot 7420 + 13 \cdot sin(12°) \cdot 7420 - 513.1 \cdot 29 + 7269.5 \cdot 86.2 + B_{Y2} \cdot 98.7 = 0 \Leftrightarrow \quad (2.3)$$

$$B_{Y2} = -4475.0N \quad (2.4)$$





The absolute displacement from the same node across all simulations was gathered and registered in table 2.1 and graphics for each element type were created as shown in figure 2.9.

As seen in the results, both quadratic TET and wedge elements quickly converge on a solution. However, the C3D6 mesh has not yet converged with 30852 elements and the C3D4 elements suffer from shear locking, therefore presenting an artificially high stiffness.

Despite the acceptable results presented by the C3D15 elements – due to the high amount of partitions needed to create the mesh – the C3D10 elements will be considered for this model. The mesh size used in the analysis was defined at 18776 elements and 30611 nodes, since it allows for good simulations results and quick computational times.

| ELASTIC DISPLACEMENT [U, mm] | | | | | |
|---|---|---|---|---|---|
| **Nº Elements** | **4 577** | **6 977** | **12 096** | **18 776** | **30 852** |
| C3D10 | 1.667 | 1.692 | 1.704 | 1.711 | 1.717 |
| C3D4 | 0.836 | 1.012 | 1.215 | 1.314 | 1.365 |
| **Nº Elements** | **###** | **###** | **12 186** | **17 577** | **30 037** |
| C3D15 | | | 1.717 | 1.720 | 1.725 |
| C3D6 | | | 1.561 | 1.604 | 1.647 |

Table 2.1: Mesh convergence study data

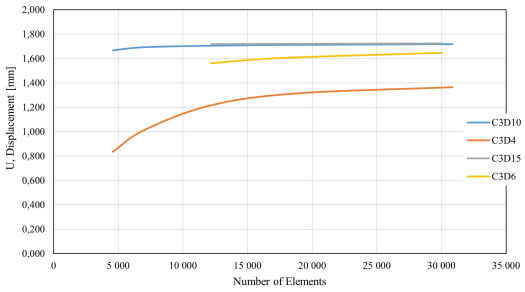



Figure 2.9: Mesh convergence graphic





is the most strained, but the far right region has also hardened. The results conclude the material reached its ultimate strength in both the sections discussed, however the value was only surpassed by 7%, meaning the base wouldn't succumb to critical failure under normal working loads.

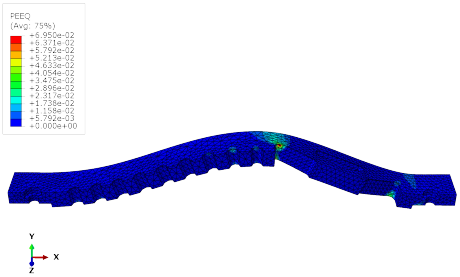



Figure 3.4: Nonlinear elastoplastic equivalent plastic strain field

### 3.3.2 Temporal Convergence Study

To comprehend how the step time influenced results, the maximum iteration time was varied across four different jobs, obtaining the results shown in graphic 3.5. Each line is associated to a maximum iteration time as represented in the graph's legend.

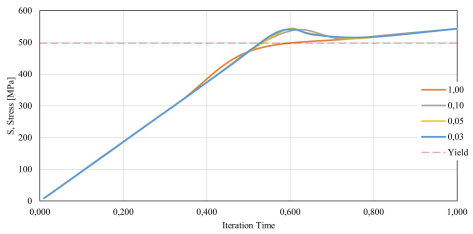



Figure 3.5: Temporal convergence study





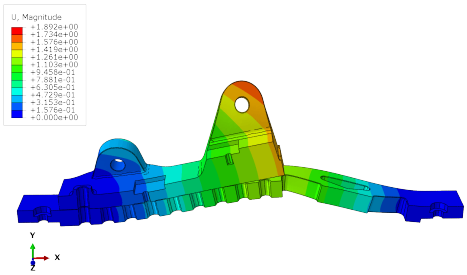



Figure 4.3: Nonlinear contact model displacement field (mm)

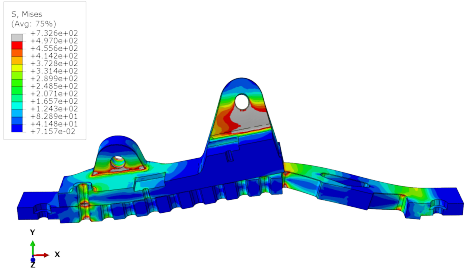



Figure 4.4: Nonlinear contact model stress field (MPa) - inclined view